a Material theory

Stress equation: $\boldsymbol{\sigma}$ is a function of $\boldsymbol{h}, \boldsymbol{\varepsilon}$

Evolution equations: $\dot{h}_{\boldsymbol{n}}$ is a function of $\boldsymbol{h}, \boldsymbol{\varepsilon}$

$\boldsymbol{\sigma}$ Stress components

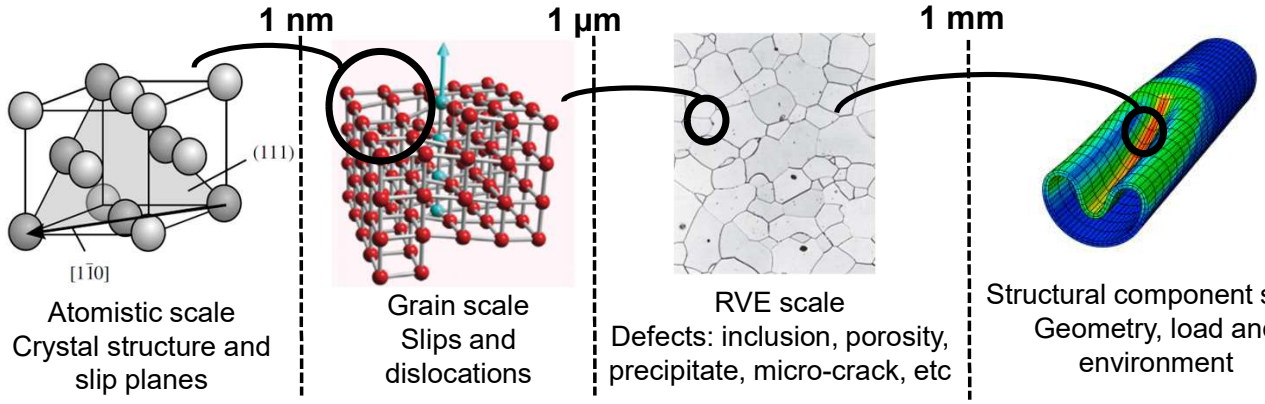
$\boldsymbol{\varepsilon}$ Strain components

$\boldsymbol{h}$ Internal state variables (ISV) e.g., hardening variables

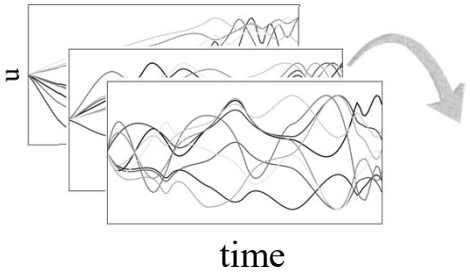
n ISV number

f and g are a *priori* unknown functions

b Material length scale

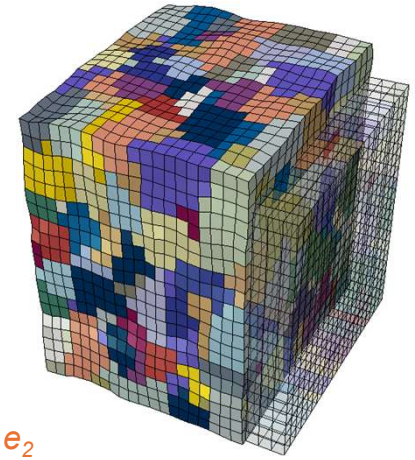


c Surrogate modelling



Loading trajectories

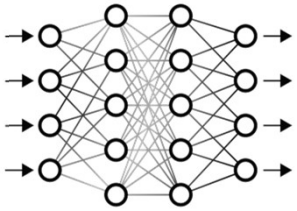
Meso-scale



$\{\bar{\boldsymbol{\varepsilon}}, \bar{\boldsymbol{\sigma}}\}$

Surrogate model

User-Material



$$d\mathbf{y} = MLP(\boldsymbol{y}, \dot{\boldsymbol{\varepsilon}})d\boldsymbol{\varepsilon}$$

$$\boldsymbol{y} = [\boldsymbol{\sigma}, \boldsymbol{h}]$$

MLP: multi-layer perceptron

Macro-scale

$\boldsymbol{\sigma}^{\Omega}$ $\mathbb{C}^{\Omega}$

$\boldsymbol{\varepsilon}^{\Omega}$, $\Delta\boldsymbol{\varepsilon}^{\Omega}$

